

AgentSight

Production Runtime Security Sensor
OS-Level Security Monitoring for AI Agents

5-Priority Implementation
Status: Production Ready

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Executive Summary

AgentSight has evolved from a simulation-based prototype to a **production-grade runtime security sensor** for AI agents. All five strategic priorities have been implemented, integrated, tested, and deployed.

5-Priority Status

Priority	Component	Status
1: Real eBPF	ebpf_loader.py	■
2: Kernel Events	probe.c	■
3: Restructure	src/runtime/	■
4: Persistence	persistence.py	■
5: Policy Engine	policy_engine.py	■

Priority 1: Real eBPF Chain

EBPFCompiler compiles C → BPF with CO-RE support. EBPFProbeRuntime manages full lifecycle. System checks for kernel version, /sys/fs/bpf, CAP_BPF.

Priority 2: Extended Kernel Events

Type	Description
EVENT_EXEC	Process execution
EVENT_EXIT	Process termination
EVENT_OPEN_FILE	File operations
EVENT_CONNECT	Network connections

Priority 3: Code Restructure

Separated src/runtime/ (production), src/ebpf/ (kernel), src/demo/ (simulation). Clear boundaries between layers.

Priority 4: Persistence + Streaming

EventStore: SQLite3 with events, sessions, security_alerts tables. Indexed for fast queries.

EventStreamer: Pluggable handlers (log, HTTP, Kafka-ready).

Priority 5: Policy Engine

Features: YAML-based rules, single/multi-event conditions, risk scoring, agent allowlisting.

Operators: eq, regex, in, contains, gt, lt.

Unified Orchestrator

AgentSightRuntime: Integrates all 5 priorities. Pipeline: Kernel → eBPF → Ring Buffer → SQLite → Policy → Alerts → Streaming.

Validation & Deployment

Tests: 5/5 passing (eBPF, Persistence, Policy, Orchestrator, Integration).

Status: Production Ready (requires vmlinux.h + CAP_BPF).